

What if I actually Run in a Running game: A novel Seated Walking-in-Place Locomotion Method in Immersive VR Running Games

Jiashu Sun

School of Electrical Engineering and Computer Science
KTH Royal Institute of Technology
Stockholm, Sweden
jiashus@kth.se

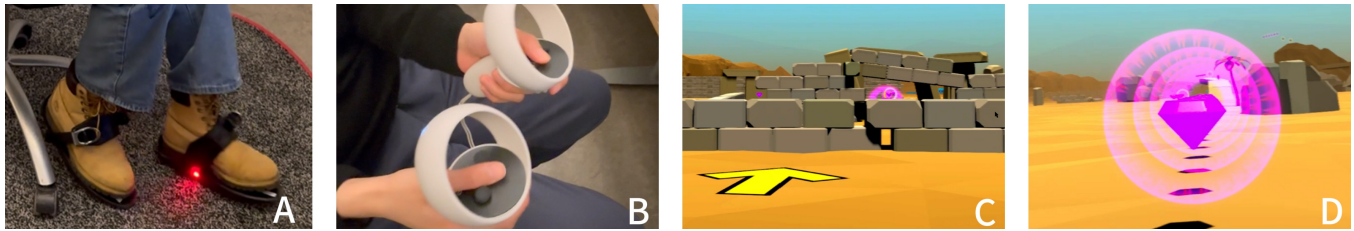


Figure 1: Gameplay: Choose either Cybershoes (A) or the hand controller (B) to start walking, navigate obstacles and run toward the finish line (C). Collect as many diamonds as possible along the way (D).

Abstract

With the development of VR technologies, the integration of physical movement into immersive experiences has become increasingly critical for enhancing user engagement and presence. As a simple yet widely adopted genre, running games are particularly sensitive to the choice of locomotion techniques, which shape the overall gameplay experience. In this study, we developed an interactive interface for a VR parkour game and conducted a comparative evaluation between a novel physical locomotion method and traditional approach. Our findings reveal key differences in user immersion, presence, and interaction, offering valuable insights for the future design and development of VR games that leverage embodied movement.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Empirical studies in HCI*.

Keywords

Virtual reality, Locomotion, Game, foot-based interaction

ACM Reference Format:

Jiashu Sun. 2025. What if I actually Run in a Running game: A novel Seated Walking-in-Place Locomotion Method in Immersive VR Running Games. In *The 38th Annual ACM Symposium on User Interface Software and Technology*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758355>

(UIST Adjunct '25), September 28–October 01, 2025, Busan, Republic of Korea.
ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758355>

1 Introduction

Running games have remained one of the most popular and accessible genres on mobile platforms since the advent of smartphones [8–10]. Their appeal lies in simple yet engaging interaction loops, rapid progression, and adaptability to both entertainment and serious applications [1, 7]. With the increasing accessibility and affordability of consumer-grade Virtual Reality (VR) head-mounted displays (HMDs) [3, 4], there is growing interest in reimagining the running game experience within immersive VR environments.

In VR, locomotion—the mechanism by which users move through virtual space—is a fundamental interaction modality [3]. While often relegated to a secondary role in complex narrative or puzzle-based games, locomotion becomes a core gameplay component in running games [11], where continuous movement is essential. Consequently, the choice of locomotion method plays a crucial role in shaping users' sense of control, presence, and overall immersion.

Previous research has investigated various VR travel techniques, ranging from teleportation and joystick-based movement to more embodied solutions. Among these, walking-in-place (WIP) methods have shown promise by offering an effective balance between immersion, comfort, and spatial efficiency [5, 6]. However, most WIP techniques remain at the research prototype stage and lack widespread adoption on consumer platforms [2]. While some preliminary work has explored specific WIP hardware devices, these studies often lack robust qualitative and quantitative evaluations of user experience.

To further explore this space and address this gap, we developed a simplified base locomotion interface embedded in a VR parkour-style running game. The goal was to assess immersion through a combination of objective performance data and subjective user

feedback. This study aims to identify key factors influencing users' sense of immersion and provide actionable insights for the design and development of future embodied locomotion systems in VR.

2 IMPLEMENTATION

Locomotion Interface Design. We developed a dual locomotion interface to support both Cybershoes-based foot input and conventional handheld controller movement within the same VR environment. Cybershoes are commercial wearable devices designed for seated walking-in-place (WIP) interaction. Each shoe is equipped with a set of inertial measurement units (IMUs) that capture the user's foot sliding motion while seated. The forward or backward movement of each foot is translated into continuous directional input through real-time signal processing. The system calculates stride frequency and intensity to dynamically adjust the player's virtual velocity, offering a natural sense of acceleration and deceleration. This allows users to simulate walking or running while minimizing whole-body fatigue.

In parallel, we implemented a traditional thumbstick-based locomotion mode using the Oculus Quest 2 hand controllers. This method relies on analog input from the controller to move the avatar, providing a baseline for comparison. Both locomotion modes were integrated using Unity's XR Interaction Toolkit to ensure consistency in character movement, collision detection, and interaction logic. **System Integration and Environment.** The VR environ-



Figure 2: Components of Cybershoes

ment was rendered through the Oculus Quest 2 headset, which provided 6DoF tracking, stereoscopic visuals, and spatial audio to ensure immersive feedback. We created a desert-themed running game using Unity, featuring a linear but slightly curved path flanked by stylized terrain elements, such as sand dunes, cacti, and ancient ruins. Along the path, interactive game elements like scattered diamonds and small obstacles (e.g., rocks and gaps) were included to encourage physical engagement and decision-making during navigation.

The experiment took place in a $1\text{m} \times 1\text{m}$ circular test area where participants remained seated on a swivel chair. The game supported both locomotion modes and automatically switched control input based on the selected mode.

3 USER STUDY AND RESULTS

Participants and Method Thirteen participants were recruited for a within-subject study to compare the new interface with traditional hand-controller locomotion in a VR running game. Each participant experienced both modes in randomized order. After each session,

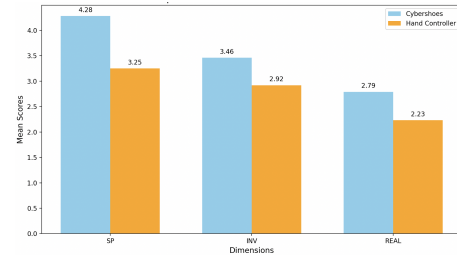


Figure 3: Comparison of Mean IPQ Subscale Scores Between Locomotion Methods

they completed the IPQ (Igroup Presence Questionnaire) and SSQ (Simulator Sickness Questionnaire), followed by semi-structured interviews. Quantitative data were analyzed using paired samples t-tests, and qualitative responses were open-coded.

Results The results showed that the new interface significantly increased overall IPQ scores (Mean = 10.53) compared to hand controllers (Mean = 8.40), with a mean difference of 2.13 ($p = 0.003$). Spatial Presence improved most notably (Mean diff = 1.03, $p < 0.001$, $d = 0.86$), followed by Involvement (0.54, $p = 0.040$, $d = 0.64$). Realism also improved (0.56, $d = 0.52$) but was not statistically significant ($p = 0.088$). Simulator sickness scores showed a slight, non-significant reduction with the new interface ($M = 1064.06$ vs. $M = 1090.15$), indicating minimal impact. SSQ scores showed no significant difference between conditions, and no meaningful correlation was found between simulator sickness and immersion.

Qualitative Insights and Future Directions Participants frequently described the foot-based motion as natural and immersive, noting that the added physical effort contributed to a stronger sense of realism. However, some reported that fatigue or unintended collisions reduced focus. Users expressed strong interest in applying the new interface to fitness-oriented scenarios and suggested enhancing interaction capabilities (e.g., jumping or crouching) to increase gameplay depth.

These insights point toward three main areas for future work: (1) developing fitness-focused applications with features like step tracking and customizable workouts; (2) improving compatibility through standardized APIs and flexible settings to support more VR game types; and (3) refining ergonomic and embodied design to further align physical input with virtual feedback and deepen immersion.

4 Conclusion

This study explored a novel walk-in-place locomotion interface for VR running games and found that it significantly enhances perceived immersion compared to traditional hand-controller methods. Through both quantitative and qualitative analysis, we demonstrated that natural foot-based input improves spatial presence and involvement. While challenges such as fatigue and interaction limitations remain, our findings highlight the potential of embodied interfaces for more immersive and physically engaging VR experiences. Future research should further explore fitness applications, broader compatibility, and expanded interaction capabilities to maximize the impact of such systems.

References

- [1] Diego Roberto Antunes and Janaine Daiane Rodrigues. 2021. Endless running game to support sign language learning by deaf children. In *International Conference on Human-Computer Interaction*. Springer, 25–40.
- [2] Costas Boletsis and Jarl Erik Cedergrén. 2019. VR locomotion in the new era of virtual reality: an empirical comparison of prevalent techniques. *Advances in Human-Computer Interaction* 2019, 1 (2019), 7420781.
- [3] Liwei Chan, Tzu-Wei Mi, Zhung Hao Hsueh, Yi-Ci Huang, and Ming Yun Hsu. 2024. Seated-WIP: Enabling Walking-in-Place Locomotion for Stationary Chairs in Confined Spaces. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–13.
- [4] Charlene Jennett, Anna L Cox, Paul Cairns, Samira Dhoparee, Andrew Epps, Tim Tijs, and Alison Walton. 2008. Measuring and defining the experience of immersion in games. *International journal of human-computer studies* 66, 9 (2008), 641–661.
- [5] Maud Marchal, Julien Pettré, and Anatole Lécuyer. 2011. Joyman: A human-scale joystick for navigating in virtual worlds. In *2011 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, 19–26.
- [6] Jesus Mayor, Laura Raya, and Alberto Sanchez. 2019. A comparative study of virtual reality methods of interaction and locomotion based on presence, cybersickness, and usability. *IEEE Transactions on Emerging Topics in Computing* 9, 3 (2019), 1542–1553.
- [7] Shawn Nikkila, Gaurav Patel, Hari Sundaram, Aisling Kelliher, and Ashutosh Sabharwal. 2012. Wind runners: designing a game to encourage medical adherence for children with asthma. In *CHI’12 Extended Abstracts on Human Factors in Computing Systems*. 2411–2416.
- [8] Mel Slater. 2002. Presence and The Sixth Sense. *Presence* 11 (08 2002), 435–439. doi:10.1162/105474602760204327
- [9] Purva Tekade, Bhavika Thakre, Kanaka Golwalkar, Anshuli Nikhare, and Darshan Surte. 2021. Roventure-An Endless Runner Game. *Journal of Examples* (2021).
- [10] Joy Van Baren. 2004. Measuring presence: A guide to current measurement approaches. *Deliverable of the OmniPres project IST-2001-39237* (2004).
- [11] Daniel Zielasko and Bernhard E Riecke. 2024. Sitting or Standing in VR: About Comfort, Conflicts, and Hazards. *IEEE Computer Graphics and Applications* 44, 2 (2024), 81–88.